

Java Programming Assignment

Abstract Classes & Method Implementation

★ Intermediate Level

■ 45 – 60 Minutes

■ Core Java / OOP

■ Problem Statement

Design a Vehicle Management System

A transport company manages different types of vehicles — **Car**, **Bike**, and **Truck**. Each vehicle shares common properties but calculates fuel efficiency and produces sounds differently. Your task is to model this system using Java **Abstract Classes**.

■ Task Instructions

Step 1 — Create Abstract Class **Vehicle**

- Instance variables: **brand**, **speed**, **fuelCapacity**
- Constructor to initialize all three variables
- Concrete method: **displayInfo()** — prints brand and speed
- Abstract method: **calculateFuelEfficiency()** — returns **double**
- Abstract method: **makeSound()** — returns **String**

Step 2 — Create 3 Concrete Subclasses

Class	Fuel Efficiency Formula	makeSound() Output
Car	$\text{fuelCapacity} / \text{speed} * 10$	"Vroom Vroom"
Bike	$\text{fuelCapacity} / \text{speed} * 15$	"Vroom Vroom"
Truck	$\text{fuelCapacity} / \text{speed} * 5$	"Horn Blows!"

Step 3 — Create **Main** Class

- Create one object each of **Car**, **Bike**, and **Truck**
- Call **displayInfo()**, **calculateFuelEfficiency()**, and **makeSound()** on each object
- Print formatted output to the console

■ Expected Output

```
Brand: Toyota | Speed: 80 km/h
Fuel Efficiency : 6.25 km/l
Sound : Vroom Vroom
```

```
Brand: Royal Enfield | Speed: 60 km/h
Fuel Efficiency : 7.50 km/l
```

Sound : Vroom Vroom

Brand: Tata Truck | Speed: 40 km/h

Fuel Efficiency : 3.75 km/l

Sound : Horn Blows!

■ Bonus Challenge

- ★ Add an abstract method **getVehicleType()** returning a String in the Vehicle class
- ★ Create a new subclass **ElectricCar** with a fuel efficiency formula of your choice
- ★ Use an **ArrayList<Vehicle>** and iterate using an enhanced for loop to print all vehicle details

■ Concepts Covered

abstract class	abstract methods	inheritance	method overriding	polymorphism	constructors	ArrayList
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